Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/30/2005
PI Name:	Leff, Laura Ph.D.		
Project Title:	The Effects of Microgravity on the Ecology of Biofilm Bacteria		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Evolutionary biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	07/01/2001	**End Date:**	06/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 2-1497		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Throughout this study several types of methods will be employed to assess bacterial community composition and gauge the responses of this community. Several molecular methods are available for our use: sequencing of DNA, PCR and subsequent analysis by denaturing gradient gel electrophoresis, and fluorescent in situ hybridization. Each allows us to characterize bacterial communities using culture-independent methods. In addition to these molecular methods, we also will rely on direct enumeration using epifluorescent microscopy and cultivation. Methods will be applied in varying combinations to material from three basic sources: 1) bacteria from the Johnson Space Center (JSC) culture collection, 2) bacterial samples collected from space shuttle/station water systems, and 3) samples collected during our ground based experimentation. By applying methods in combination, we can understand the ecology of microbes in this system. To examine responses of bacteria to simulated microgravity and study biofilm formation by bacteria from space station		

water systems.

Formation of biofilms in water systems may enhance bacterial survival and increase the potential for negative impacts on water quality.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Bacteria are chronic inhabitants of water systems, including those associated with NASA activities. There are two primary objectives for this study. First, we are using state of the art molecular methods to assess bacterial community composition in water systems. This includes use of 16S rDNA sequencing and subsequent phylogenetic analysis to identify and characterize bacteria isolated from various systems (including space shuttle water systems, bacteria previously isolated from Mir, etc.). We will use our molecular ecology expertise to examine bacteria that defy cultivation (bacteria under oligotrophic conditions, like in pure water, are typically difficult to culture and thus other methods must be employed to examine this important group). Second, using bacteria isolated previously from the JSC collection and any new isolates, we are conducting ground based experiments on factors that affect biofilm formation and particularly the influence of microgravity on the growth and success of bacteria. This will be accomplished using slow turning lateral vessels to simulate microgravity. The overall significance is that this will provide the basic information needed to understand factors that influence the number and type of bacteria found in shuttle and space station water systems.

Task Progress:

Summaries of completed sections of the project are described below and include papers that were published this year and one that is currently in review. In addition, on going experiments are focused on examination of long term survival in water (one manuscript on this topic is in review and another is in preparation) and exploring the mechanisms behind population level responses observed. In the latter case, we are conducting experiments using microarrays to examine changes in gene expression under reduced gravity. This will lead to insights as to the basis of the phenomena described below.
In Baker et al. (2004) Escherchia coli was subjected to different rotation speeds during clino-rotation and significant differences between simulated reduced gravity and normal gravity controls were observed at higher speeds (30-50 rpm). There was no apparent affect of removing samples on the results obtained. When E. coli was grown in minimal medium (at 40 rpm), cell size was not affected by modeled reduced gravity and there were few differences in cell numbers. However, in higher nutrient conditions (i.e., dilute nutrient broth), total cell numbers were higher and cells were smaller under reduced gravity compared to normal gravity controls. Overall, the responses to simulated reduced gravity varied with nutrient conditions; larger surface to volume ratios may help compensate for the zone of nutrient depletion around the cells under reduced gravity.

In the article by Baker and Leff (2005, In press), four bacterial isolates (Chryseobacterium sp., Pseudomonas fluorescens and two Stenotrophomonas maltophilia isolates) from the water system aboard the Mir space station were grown in nutrient broth under modeled reduced gravity, that was simulated using clino-rotation. Planktonic and sessile cells were sampled over a 7-day period and enumerated using the Live/Dead BacLight kit and DAPI. On some of the sampling days for all the isolates, planktonic cell counts were higher under modeled reduced gravity compared to the normal gravity controls. In contrast, the number of cells of P. fluorescens and one S. maltophilia isolate attached to stainless steel disks was higher under modeled reduced gravity compared to normal gravity, whereas for Chryseobacterium sp. and the other S. maltophilia isolate no such differences were observed. Differences among isolates in hydrophobicity and motility appeared to influence the growth of planktonic cells under modeled reduced gravity but did not appear to be related to biofilm formation.

Song and Leff (2005, In press) report on the identification of 20 bacterial isolates (supplied by NASA) from the Mir space station water system based on the Vitek GNI+ test card, API 20NE, and 16S rRNA gene sequencing. The identification of only 1 isolate agreed among the 3 techniques. The utility of the API 20NE and Vitek GNI+ test card approaches for identifying these isolates was limited. Although 16S rRNA gene sequencing effectively identified many of the bacteria to the genus level, 74% of the isolates could not be identified to the species level. Isolates were also characterized based on motility and hydrophobicity. About 40% of the isolates were motile and four isolates were hydrophobic, suggesting that many of the bacteria have the potential to colonize surfaces and form biofilms. These findings demonstrate the difficulties in identifying bacteria from some environments to the species level and have implications for determining the risks of contamination in water systems of space shuttles and stations.

In Baker and Leff (2005) responses to modeled reduced gravity of space station were compared between water system bacterial isolates to other isolates of the same species. Bacterial isolates, Stenotrophomonas paucimobilis and Acinetobacter radioresistens, originally recovered from the water supply aboard the International Space Station (ISS) were grown in nutrient broth under modeled reduced gravity. Their growth was compared to type strains S. paucimobilis ATCC 10829 and A. radioresistens ATCC 49000. A. radioresistens ATCC 49000 and the two ISS isolates showed similar growth profiles under modeled reduced gravity compared to normal gravity, whereas S. paucimobilis ATCC 10829 was negatively affected by modeled reduced gravity. These results suggest that microgravity might have selected for bacteria that were able to thrive under this unusual condition. These responses, coupled with impacts of other features (such as radiation resistance and ability to persist under very oligotrophic conditions), may contribute to the success of these water system bacteria.

In a paper (Song and Leff, in review) that is under consideration for publication, the impacts of Mg2+ on biofilm formation were examined because Mg2+ can potentially influence bacterial adhesion directly through effects on electrostatic interactions and indirectly by affecting physiology-dependent attachment processes. In this particular study, Pseudomonas fluorescens (from the Mir) was used to investigate the influence of Mg2+ concentration (0, 0.1 and 1.0mM MgCl2) on biofilm growth. Planktonic and attached cells were enumerated (based on DAPI staining) while biofilm structures were examined via confocal laser scanning microscopy and three-dimensional structures were reconstructed. Mg2+ concentration had no influence on growth of planktonic cells but, during biofilm formation, Mg2+ increased the abundance of attached cells. For attached cells, the influence of Mg2+ concentration changed over time, suggesting that the role of Mg2+ in bacterial attachment is complex and dynamic. Biofilm structures were heterogeneous and surface colonization and depth increased with increasing Mg2+ concentrations. Overall, for P. fluorescens, Mg2+ increased initial attachment and altered subsequent biofilm formation and structure.

Bibliography Type:	Description: (Last Updated: 04/09/2020)
Abstracts for Journals and Proceedings	Baker PW, Leff LG. "Bacterial attachment to stainless steel disks in a simulated microgravity environment." Ohio Branch Meeting American Society for Microbiology. 2004, Youngstown, OH. Ohio Branch Meeting American Society for Microbiology. 2004. , Apr-2004
Abstracts for Journals and Proceedings	Baker PW, Meyer ML, Leff LG. "Responses of Escherchia coli ATCC 26 to simulated microgravity." American Society for Microbiology, New Orleans LA, May 2004. American Society for Microbiology, May 2004. , May-2004
Abstracts for Journals and Proceedings	Meyer ML, Kuny V, Baker PW, Leff LG. "Biofilm formation by bacteria from the Mir Space Station to stainless steel disks." American Society for Microbiology, New Orleans LA, May 2004 . American Society for Microbiology, May 2004 . , May-2004
Abstracts for Journals and Proceedings	Song B, Leff LG. "Effects of Ca+2 and Mg+2 on biofilm formation of Pseudomonas fluorescens in static conditions." American Society for Microbiology, New Orleans LA, May 2004. American Society for Microbiology, May 2004. , May-2004
Articles in Peer-reviewed Journals	Baker PW, Meyer ML, Leff LG. "Escherichia coli growth under modeled reduced gravity." Microgravity Sci Technol. 2004;15(4):39-44. PMID: 15768486 , Jan-2004
Articles in Peer-reviewed Journals	Baker PW, Leff LG. "Intra-specific differences in bacterial responses to modeled reduced gravity." J Appl Microbiol. 2005;98(5):1239-46. PMID: 15836494 , May-2005
Articles in Peer-reviewed Journals	Baker PW, Leff L. "Attachment to stainless steel by Mir space station bacteria growing under modeled reduced gravity at varying nutrient concentrations." Biofilms, 2005 Jan;2(1):1-7. , Jan-2005
Articles in Peer-reviewed Journals	Song B, Leff LG. " Identification and characterization of bacterial isolates from the Mir space station." Microbiol Res. 2005;160(2):111-7. PMID: 15881827 , Jun-2005
Articles in Peer-reviewed Journals	Song B, Leff LG. "Influence of magnesium ions on biofilm formation by Pseudomonas fluorescens." Microbiol Res. 2006 Nov;161(4):355-61. Epub 2006 Mar 6. https://doi.org/10.1016/j.micres.2006.01.004 ; PubMed PMID: 16517137 , Nov-2006
Articles in Peer-reviewed Journals	Baker PW, Leff LG. "Mir space station bacteria responses to modeled reduced gravity under starvation conditions." Adv Space Res. 2006;38(6):1152-8. Available online 21 June 2006. https://doi.org/10.1016/j.asr.2006.05.014 , Jun-2006
Dissertations and Theses	Song B. "Identification and characterization of bacterial isolates from the Mir space station and the impact of Mg2+ concentration on biofilm formation of Pseudomonas fluorescens." MS Thesis, Kent State University, May 2004. , May-2004